

# TrES-1b

R-0217

Benchmark run · workflow W8-benchmark-deep · record deep-bench\_TRES-1\_180720 · created 2026-07-17T00:57:49+00:00

## BENCHMARK: NON-DETECTION

### Results

|                                                   |                                |
|---------------------------------------------------|--------------------------------|
| Mid-Transit Time (Tmid)                           | 2458319.7468 +/- 0.002 BJD_TDB |
| Ratio of Planet to Stellar Radius (Rp/R*)         | 0.048 +/- 0.054                |
| Transit depth (Rp/Rs)^2                           | 0.23 +/- 0.52 [%]              |
| Orbital Inclination (inc)                         | 89.02 +/- 1.4                  |
| Airmass coefficient 1 (a1)                        | 0.047 +/- 0.056                |
| Airmass coefficient 2 (a2)                        | 3.0 +/- 1.7                    |
| Scatter in the residuals of the lightcurve fit is | 204.21 %                       |
| Best Comparison Star                              | #2 - [418, 142]                |
| Optimal Aperture                                  | 0.87                           |
| Optimal Annulus                                   | 4.0                            |
| Transit Duration (day)                            | 0.0927 +/- 0.0077              |

### Accuracy vs published ephemeris (ExoClock/NEA)

|                           |                                           |
|---------------------------|-------------------------------------------|
| Rp/R■ fit vs published    | 0.048 ± 0.054 vs 0.1358 (deviation 1.63σ) |
| Duration fit vs published | 0.0927 d vs 0.1042 d (deviation 1.49σ)    |
| Mid-time O–C              | 1.8 min                                   |
| Depth SNR                 | 0.9                                       |
| Residual scatter          | 204.21 %                                  |

### Light curve

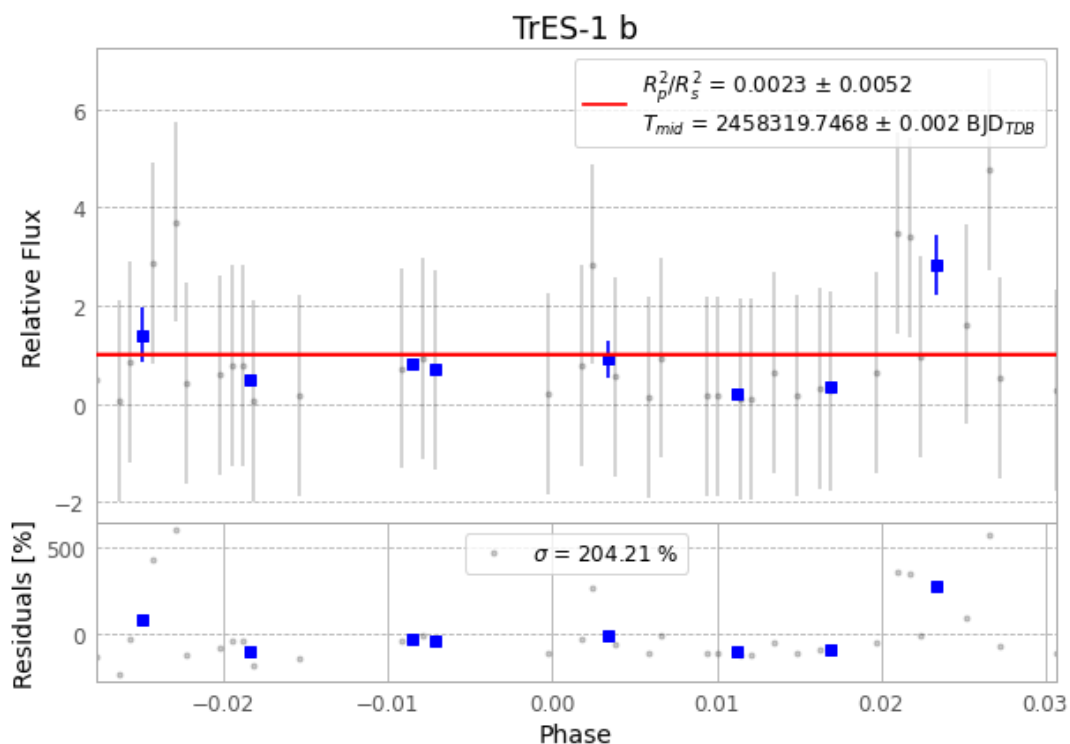


Figure 1 — FinalLightCurve\_TrES-1 b\_2018-07-19.png (SHA-256 0156069fd6acad03...)

## Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [299, 116], 3 comparison star(s). Exact configuration: parameters.inits\_json in record.json (SHA-256 759d71a637e2a2cd...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline\_commit 825eb7e

## Data provenance

90 FITS frames (59 MB), TRES-1180720034212.FITS → TRES-1180720080912.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-TRES-1-180720.log (e734fcc1d2da...), FinalLightCurve\_TrES-1 b\_2018-07-19.png (0156069fd6ac...), FinalLightCurve\_TrES-1 b\_2018-07-19.csv (578551c3e9cd...)

## Compute resources

Wall time 245.4 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

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Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-07-17 00:57Z.